

30-05-2026

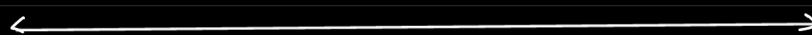
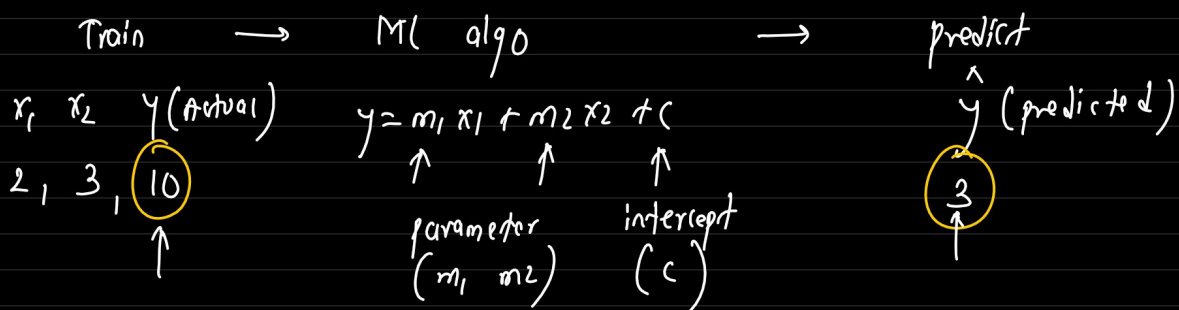
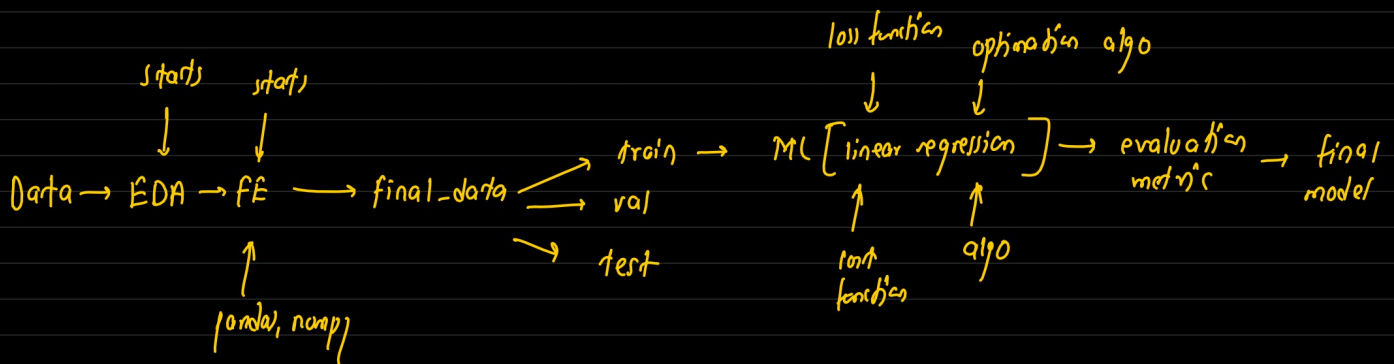
Agenda: ML-2

Topic covered:

- what is ML?
- Types of learning
- Data split concept

Today:

evaluation, loss function, cost function, [overfit, underfit, ideal model],
evaluation metrics (regression, classification), regression [linear, polynomial]



difference

→ we require a method to tell ML algo about error, so it can adjust itself.

Evaluation Metrics:

→ whether our model is learning or not? → loss function
whether our model is good or bad? → Test model perf

		<u>Mark</u>
Teacher	→ student	→ 50/100
	→ student	→ 50/100
		→ 50/100

→ $[m_{\text{new}} - m_{\text{old}} \rightarrow +ve] \rightarrow \text{formula}$
→ Test

y (Actual/real value)	$\hat{y}$ (ML predicted values)	$y - \hat{y}$ (error)
10	5	5
15	7	8
20	19	1
40	50	-10
		4

Error calculation → each data point → loss function
→ set of data (train, val) → cost function (4)

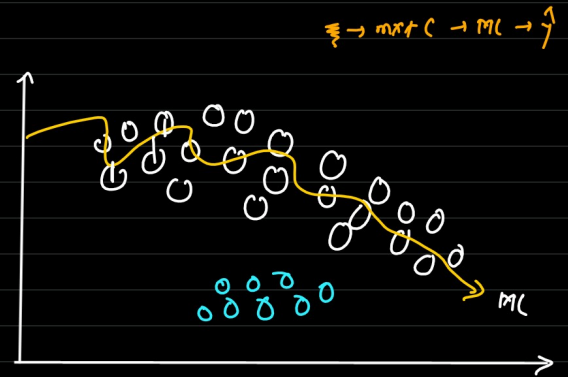
loss function: Measure the error for a single example

cost function: Sum of loss function over the dataset (train, val, test)

Train data $\rightarrow$ cost function $\rightarrow$ total error $\rightarrow 180$

Test data $\rightarrow$ $\rightarrow 400$

Train loss $\ll$ Test/validation loss
 $\rightarrow$ overfitting

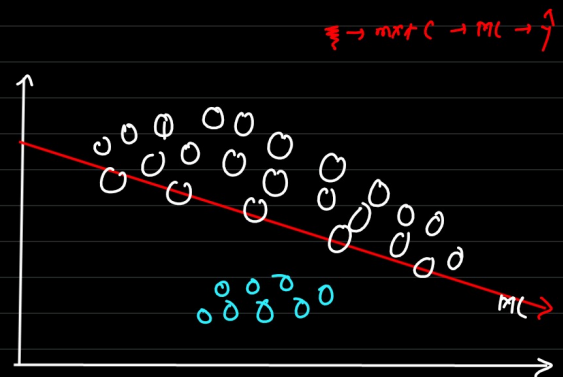


$\xi \rightarrow$ Train set

$\xi \rightarrow$ Test set

Train $\rightarrow 250$

Test $\rightarrow 230$



Train loss $\approx$ Test/valid loss $\rightarrow$ ideal case

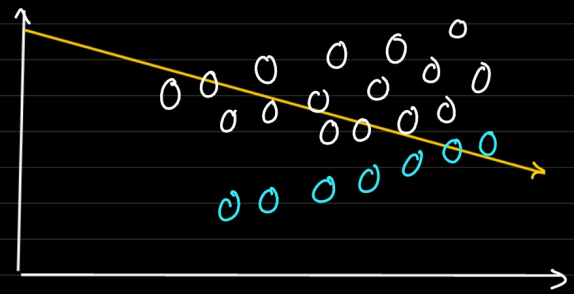
Train $\rightarrow 30$

Test $\rightarrow 32$

$\rightarrow$ good fit

Train loss & Test loss $\rightarrow$ bad
 $\rightarrow$ underfitting case

Train $\rightarrow$ high loss
 Test $\rightarrow$ high loss



$\rightarrow$ Train

$\rightarrow$ Test

	y	$\hat{y}$	
overfitting	10	10	} train $\rightarrow$ 100%
	20	20	
	30	30	
	100	30	} test $\rightarrow$ poor fit
	200	40	

	y	$\hat{y}$	
underfitting	10	1	} train $\rightarrow$ failed to learn any pattern
	20	2	
	30	3	
	100	10	} test $\rightarrow$ failed
	200	20	

	y	$\hat{y}$	
good fitting	10	9	} train $\rightarrow$ ideal fit
	20	16	
	30	28	
	100	92	} test
	200	214	

Evaluation metrics :-

(Regression)

$$(1) \sum_{i=1}^n (y - \hat{y}) \quad (2) \frac{1}{n} \sum_{i=1}^n (y - \hat{y})$$

$$(3) \text{MSE} \rightarrow \frac{1}{n} \sum_{i=1}^n (y - \hat{y})^2$$

Mean Square Error

$y \quad \hat{y} \rightarrow$ height
 10 $\rightarrow$ 8 $\rightarrow 2^2 \rightarrow 4$
 20 $\rightarrow$ 14 $\rightarrow 6^2 \rightarrow 36 \rightarrow$
 30 $\rightarrow$ 40 $\rightarrow 10^2 \rightarrow 100 \rightarrow$

value $^2 \rightarrow$ small

$\downarrow$ increase

value $\rightarrow$ bigger

$$(4) \text{ MAE} \rightarrow \frac{1}{n} \sum_{i=1}^n |y - \hat{y}|$$

Mean Absolute Error

→ same unit as target variable

$$(5) \text{ RMSE} \rightarrow \sqrt{\frac{1}{n} \sum_{i=1}^n (y - \hat{y})^2}$$

Root mean square error

(6) Modified MSE → cost function →

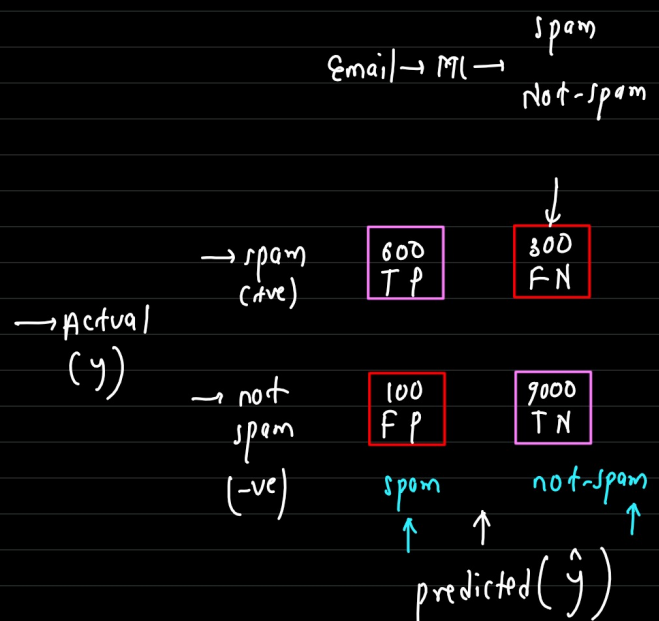
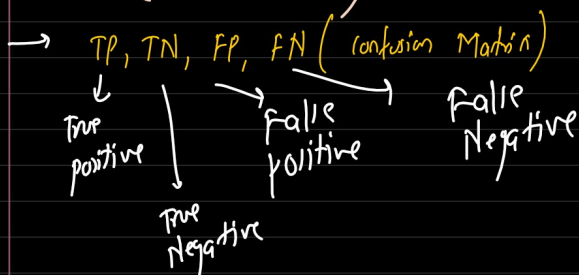
$$\rightarrow \frac{1}{2n} \sum_{i=1}^n (y - \hat{y})^2$$

$$(1) \frac{1}{n} \sum_{i=1}^n (y - \hat{y})^2 \rightarrow \text{derivate} \rightarrow \frac{2}{n} (y - \hat{y}) \cdot \frac{\partial}{\partial \theta} (y - \hat{y})$$

$$x^n \rightarrow n x^{n-1}$$

$$(2) \frac{1}{2n} \sum_{i=1}^n (y - \hat{y})^2 \rightarrow \frac{1}{2} x^2 (y - \hat{y}) \frac{d}{d\theta} (y - \hat{y})$$

Evaluation Metrics
(classification)



Truly classified as positive $\rightarrow$ TP
 Truly classified as Negative $\rightarrow$ TN
 False classified as Negative $\rightarrow$ FN
 False classified as positive $\rightarrow$ FP

		↓ A	↓ B	↓ C	↓ D
Actual	$\rightarrow$ A	☐	☐	☐	☐
	$\rightarrow$ B	☐	☐	☐	☐
	$\rightarrow$ C	☐	☐	☐	☐
	$\rightarrow$ D	☐	☐	☐	☐

$$\text{Accuracy: } \frac{(TP + TN)}{(TP + TN + FP + FN)}$$

(All true cases) / (All cases)

: overall correctness

	Actual	pred	
	Spam	Spam	100%
	Spam	Not-spam	0%
	Not-spam	Not-spam	100%
	Not-spam	Spam	

precision:

: How often we are correct

How many times we are actually correct (true)

Total times we predict correct (true)

$$= \frac{(TP)}{(TP + FP)}$$

$$\frac{\text{Actual positive}}{\text{Total predicted positive}} = \frac{600}{700}$$

$\rightarrow$ Spam

can/cv $\rightarrow$ +ve
non $\rightarrow$ -ve

$\rightarrow$ cancer (true)	TP	FN
$\rightarrow$ Actual (y)		
$\rightarrow$ not (-ve)	FP	TN
	cancer	not-cancer
	↑	↑
	predicted ($\hat{y}$)	

Recall: How many actual spam caught

$$= \frac{\text{caught spam}}{\text{Total Actual spam}} = \frac{600}{600 + 300}$$

F1 score: Harmonic mean $\rightarrow$ precision & recall

$$= \frac{2 (\text{precision} \times \text{Recall})}{\text{precision} + \text{recall}}$$

